Bayesian inference of stochastic phase dynamics in limit-cycle oscillators

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I. INTRODUCTION

Literature survey

- Theory of noisy oscillators
  - 2008, Phase reduction of stochastic limit cycle oscillators[1]
  - 2008, Comment on" Phase Reduction of Stochastic Limit Cycle Oscillators"[2]
  - 2009, Stochastic phase reduction for a general class of noisy limit cycle oscillators[3]
  - 2014, Asymptotic phase for stochastic oscillators[4]
  - 2023, A universal description of stochastic oscillators[5]
- Data analysis methods
  - 2017, *Neural Comput.*, Time series decomposition into oscillation components and phase estimation[6]
  - 2021, *ELife*, A state space modeling approach to real-time phase estimation[7]
  - 2021, *Sci. Rep.*, Real-time estimation of phase and amplitude with application to neural data[8]
  - 2023, *Sci. Rep.*, An extended Hilbert transform method for reconstructing the phase from an oscillatory signal[9]

II. GENERATIVE MODEL AND INFERENCE OF STOCHASTIC OSCILLATIONS

Many physical, chemical, and biological systems exhibit approximately periodic dynamics that are continually perturbed by fluctuations. Examples include biochemical oscillators subject to

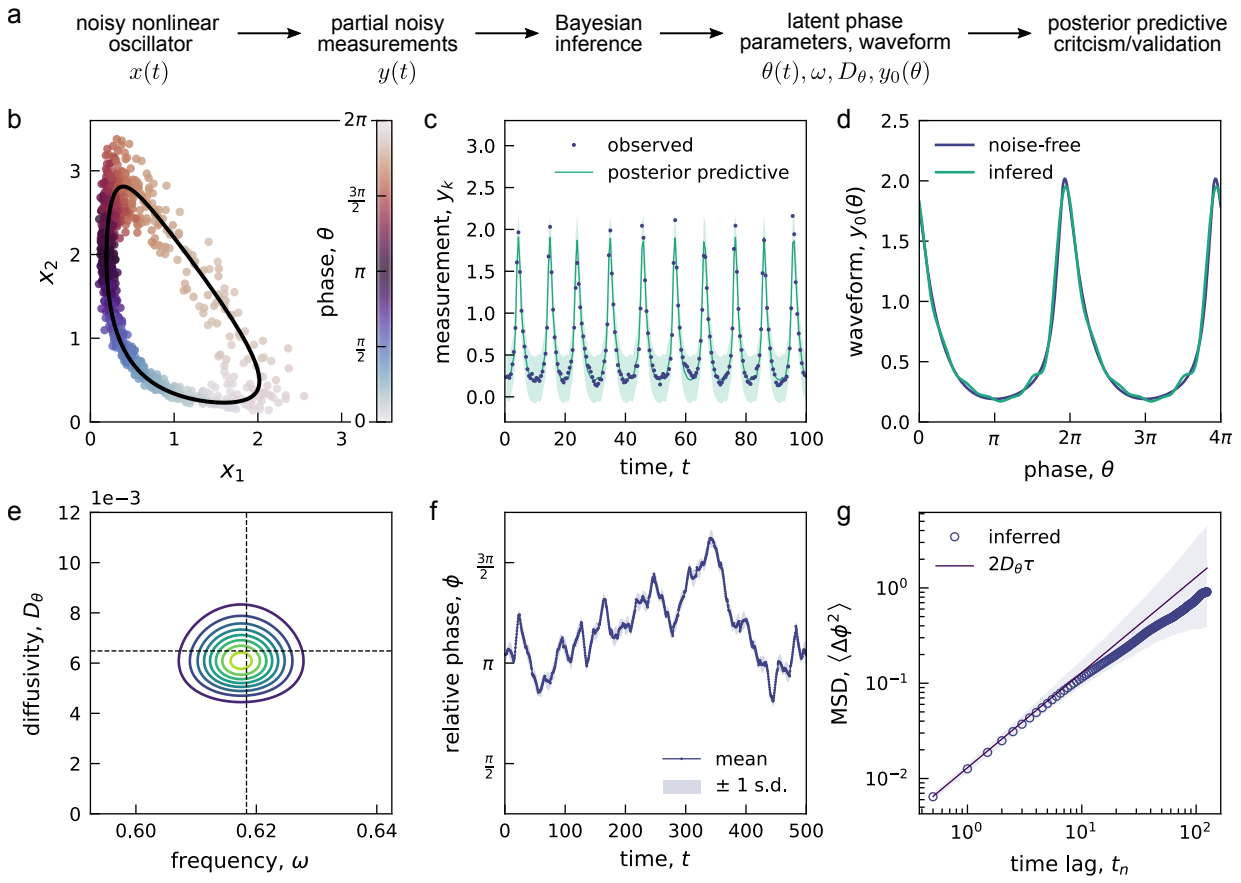


FIG. 1. **Bayesian inference of stochastic phase dynamics from partial noisy observations.** (a) Schematic overview of the inference pipeline. Noisy oscillatory dynamics are observed through partial noisy measurements, from which Bayesian inference estimates the latent phase trajectory, observation waveform, and stochastic phase dynamics. The resulting reduced model is validated through posterior predictive checks and dynamical statistics. (b) Noisy trajectory of the glycolytic oscillator in state space (x_1, x_2). The deterministic limit cycle is shown in black, and trajectory points are colored by phase θ. (c) Observed time series (points) and posterior predictive distribution (green band) generated from the inferred phase model. (d) Observation waveform $y_0(\theta)$ as a function of phase. The inferred waveform reproduces the noise-free limit-cycle observable. (e) Posterior distribution over the frequency ω and phase diffusivity D_θ. Dashed lines indicate values predicted by stochastic phase reduction of the underlying dynamical system. (f) Relative phase $\phi(t) = \theta(t) - \hat{\omega}t$, showing stochastic phase wandering after subtraction of the mean rotation. The shaded region indicates one posterior standard deviation. (g) Drift-subtracted phase mean-squared displacement, $\langle\Delta\phi^2(\tau)\rangle$, computed from posterior phase samples. The solid line shows the diffusive prediction $2D_\theta\tau$ using the inferred phase diffusivity.

molecular noise, active colloids driven by stochastic forces, and physiological rhythms influenced by environmental variability REFS. Our goal is to infer the characteristic frequency and stochastic phase dynamics of such systems directly from noisy observations.

A. Generative model for noisy oscillations

We consider an underlying dynamical system whose state is described by a vector $x(t)$ evolving in a d-dimensional state space. In the absence of fluctuations, the dynamics evolve according to a deterministic vector field $f(x)$ that admits a stable limit cycle $x_0(t)$. To account for stochastic fluctuations, we assume that the state is driven by additive noise,

$$\dot{x} = f(x) + G\,\xi(t), \tag{1}$$

where $\xi(t) \in \mathbb{R}^d$ is a vector of independent zero-mean white-noise processes satisfying

$$\langle \xi(t) \rangle = 0, \qquad \langle \xi(t)\xi(t')^\top \rangle = I\delta(t-t'). \tag{2}$$

The constant matrix $G \in \mathbb{R}^{d\times d}$ specifies the magnitude and correlations of fluctuations in state space. Throughout this work, we assume that the noise is uncorrelated in time and sufficiently weak that trajectories remain close to the deterministic limit cycle. While these assumptions can be relaxed, they provide a transparent starting point for developing the inference framework.

In many applications, the full state $x(t)$ is not directly observable. Instead, experiments provide observations

$$y(t) = h\,(x(t)) + \eta(t), \tag{3}$$

where $y(t) \in \mathbb{R}^q$ denotes the observed signal, $h(x)$ is an observation function, and $\eta(t)$ represents measurement noise. The observation model accommodates both partial observations of the state and nonlinear transformations of the underlying dynamics. For example, $h(x)$ may correspond to a single state variable, a subset of variables, or a more complicated experimentally measured quantity that depends on the underlying state. We model the measurement noise $\eta(t) \in \mathbb{R}^q$ as a vector of independent zero-mean white-noise processes with covariance

$$\langle \eta(t)\eta(t')^\top \rangle = R\delta(t-t'), \tag{4}$$

where the matrix $R \in \mathbb{R}^{q\times q}$ is the measurement noise covariance. The resulting framework distinguishes between dynamical fluctuations in state space and uncertainty introduced by the observation process.

Given a time series of observations $y_{1:N} = \{y(t_1), y(t_2), \ldots, y(t_N)\}$, our objective is to infer a low-dimensional description of the oscillatory dynamics underlying the observed signal. In the next section, we show how the high-dimensional stochastic dynamics can be reduced to an effective phase description that serves as the foundation of our inference framework.

B. Reduced stochastic phase model

The state-space description introduced above provides a general framework for noisy oscillatory dynamics but is often unnecessarily complex for analyzing oscillations observed over many cycles. For a stable limit-cycle oscillator, much of the long-time behavior is captured by progression around the cycle itself. This observation motivates a reduced description in terms of a single phase variable that tracks the oscillator's position along the cycle.

Classical phase reduction theory is based on the asymptotic phase function $\Theta(x)$, which assigns a phase coordinate $\theta = \Theta(x)$ to states in the neighborhood of a stable limit cycle. The level sets of $\Theta(x)$, known as isochrons, consist of states that converge asymptotically to the same point on the cycle.[10–12] In the weak-noise limit, the dynamics can be reduced to an effective stochastic equation for the phase alone.[1,3] The resulting phase undergoes drift and diffusion characterized by an effective frequency ω and phase diffusivity D_θ. A brief summary of stochastic phase reduction and its connection to the underlying state-space dynamics is provided in the Supplementary Material.

For observations sampled at times $t_k = t_0 + k\Delta t$, we model the unwrapped phase as a discrete-time drift–diffusion process,

$$\theta_{k+1} = \theta_k + \omega\Delta t + \varepsilon_k, \tag{5}$$

where $\varepsilon_k \sim \text{Normal}\,(0, 2D_\theta\Delta t)$ is a normally distributed random variable with zero mean and variance $2D_\theta\Delta t$. The latent phase variables $\theta_{1:N}$ together with the parameters ω and D_θ define the reduced stochastic phase model.

To relate the latent phase trajectory to the observations, we restrict the observation function to the deterministic limit cycle and define the (noise-free) waveform

$$y_0(\theta) = h\left(x_0(\theta)\right), \tag{6}$$

where $x_0(\theta)$ denotes the deterministic limit cycle parameterized by the phase. The resulting observation model is

$$y_k = y_0(\theta_k) + \eta_k, \tag{7}$$

where $\eta_k \sim \mathrm{Normal}\,(0, R_{\mathrm{eff}})$ represents both measurement noise and residual variability associated with neglected amplitude fluctuations. Consequently, the effective covariance matrix R_{eff} generally differs from the true measurement noise covariance R and should be interpreted as a coarse-grained description of uncertainty in the reduced model.

In general, the waveform $y_0(\theta)$ is not known a priori and must be inferred from the data. This periodic function may correspond to a scalar signal ($q = 1$) or a multidimensional observation ($q > 1$), depending on the experimental system under consideration. Throughout this work, we represent $y_0(\theta)$ using a truncated Fourier series whose coefficients—collectively denoted β—are estimated jointly with the phase trajectory and dynamical parameters. This reduced stochastic phase model provides a low-dimensional generative description of noisy oscillations that serves as the foundation for the Bayesian inference framework developed below.

C. Bayesian phase and parameter inference

The reduced stochastic phase model defines a probabilistic relationship between the latent phase trajectory $\theta_{1:N}$, the model parameters λ, and the observations $y_{1:N}$. The parameter set $\lambda = \{\omega, D_\theta, \beta\}$ contains the effective frequency ω, phase diffusivity D_θ, and the waveform coefficients β. Given a time series of observations, our objective is to infer both the latent phase trajectory and the unknown model parameters within a Bayesian framework. Bayes' theorem expresses the posterior distribution as

$$p(\theta_{1:N}, \lambda \mid y_{1:N}) \propto p(y_{1:N} \mid \theta_{1:N}, \lambda)\, p(\theta_{1:N} \mid \lambda)\, p(\lambda), \tag{8}$$

where the likelihood $p(y_{1:N} \mid \theta_{1:N}, \lambda)$ is determined by the observation model, the phase prior $p(\theta_{1:N} \mid \lambda)$ is determined by the stochastic phase dynamics, and the parameter prior $p(\lambda)$ describes any prior knowledge of the model parameters.

For fixed parameters λ, inference of the latent phase trajectory requires evaluation of the smoothing distribution $p(\theta_{1:N} \mid y_{1:N}, \lambda)$. Standard approaches include extended Kalman filtering and smoothing, which are computationally efficient and readily accommodate multidimensional observations.[13,14] For a one-dimensional latent phase, however, it is practical to discretize the phase coordinate and directly propagate probability densities over the phase domain. This global phase filter naturally accommodates the periodic geometry of phase and avoids local linearization of the observation model. The resulting smoothing distribution provides probabilistic estimates of the latent phase trajectory together with associated uncertainty.

The hidden Markov structure of the reduced phase model also enables efficient evaluation of the marginal likelihood $p(y_{1:N} \mid \lambda)$, which quantifies the probability of the observed data under a given parameter set. In practice, this quantity is computed directly during the filtering procedure and does not require explicit enumeration of phase trajectories. Parameter inference is then performed using the posterior distribution

$$p(\lambda \mid y_{1:N}) \propto p(y_{1:N} \mid \lambda)\, p(\lambda). \tag{9}$$

In this work, we estimate the model parameters using maximum a posteriori (MAP) inference, which identifies the parameter values $\hat{\lambda}$ that maximize the posterior probability. Uncertainty in the inferred parameters is subsequently approximated using a Laplace approximation to the posterior distribution. The corresponding smoothing distribution is then used to infer the latent phase trajectory and to quantify uncertainty in the inferred dynamics. Additional details regarding the waveform parameterization, prior distributions, likelihood evaluation, global phase filter, and numerical optimization are provided in the Supplemental Material.

D. Example: glycolytic oscillator

To illustrate the inference framework, we consider a stochastic glycolytic oscillator observed through a single noisy measurement.[15] Figure 1 summarizes the complete workflow, from the underlying state-space dynamics to the inferred stochastic phase model. The oscillator evolves in a two-dimensional state space but is observed through a one-dimensional signal, providing a representative example of partial and noisy observations. The inferred phase trajectory provides a consistent phase coordinate for trajectories in the full state space (Fig. 1b). Combined with the inferred waveform, it yields a reduced model whose posterior predictive distribution reproduces the observed time series (Fig. 1c). These results demonstrate that the latent phase variable captures the dominant oscillatory dynamics despite stochastic fluctuations in the underlying state-space dynamics.

Bayesian inference simultaneously estimates the waveform $y_0(\theta)$ and the parameters governing the effective phase dynamics. The inferred waveform closely reproduces the deterministic limit-cycle observable (Fig. 1d), while the posterior distribution over ω and D_θ quantifies uncertainty in the effective frequency and phase diffusivity (Fig. 1e). The inferred parameter values agree closely with predictions obtained from stochastic phase reduction of the underlying dynamical system.

The inferred phase trajectory further enables direct characterization of phase diffusion. After subtracting the deterministic phase drift, the relative phase $\phi(t) = \theta(t) - \hat{\omega}t$ undergoes a random walk due to stochastic variations in the oscillator period (Fig. 1f). The corresponding mean-squared displacement (MSD) grows linearly with lag time (Fig. 1g), as expected for diffusive phase dynamics. Together, these results suggest that the reduced stochastic phase model provides an accurate description of both the observable waveform and the long-time phase dynamics. The theoretical basis and limitations of this correspondence are developed in the following section.

III. STOCHASTIC PHASE REDUCTION AND EFFECTIVE OBSERVATION NOISE

The reduced stochastic phase model introduced in Sec. II is motivated by stochastic phase reduction, which describes noisy limit-cycle oscillators in terms of an effective phase undergoing drift and diffusion. The purpose of this section is to clarify the interpretation of the inferred phase coordinate and explain the role of effective observation noise in recovering the correct long-time phase dynamics. These ideas are illustrated using the Stuart–Landau oscillator, a minimal model for noisy oscillations (Fig. 2).

A. Asymptotic and projection phase coordinates

The latent phase inferred by the Bayesian framework is most naturally interpreted as a projection phase θ_{p} rather than the asymptotic phase θ. This distinction is illustrated in Fig. 2a for the Stuart–Landau oscillator, for which both decompositions can be obtained analytically (see the Supplementary Material). In the asymptotic decomposition, states are parameterized by an asymptotic phase θ and an amplitude coordinate u defined relative to the isochrons. In the projection decomposition, states are associated with the nearest point on the deterministic limit cycle, yielding a projection phase θ_{p} and a transverse amplitude coordinate u_{p}. This construction arises naturally in the inference framework because observations are represented through a waveform defined on the deterministic limit cycle and do not require knowledge of the isochron geometry.

The asymptotic phase and amplitude coordinates (θ, u) associated with the noisy trajectory are shown in Fig. 2b. While the phase exhibits unbounded diffusive fluctuations, the amplitude remains bounded and fluctuates about the limit cycle. The projection coordinates $(\theta_{\mathrm{p}}, u_{\mathrm{p}})$ provide an alternative decomposition of the same trajectory and exhibit analogous behavior. In both cases, the

amplitude coordinate quantifies transverse displacements from the limit cycle, whose autocorrelation decays exponentially with a common relaxation time τ_r (Fig. 2c). More generally, this time scale is determined by the slowest relaxation mode transverse to the limit cycle and is therefore independent of the particular phase–amplitude coordinates used to describe fluctuations near the

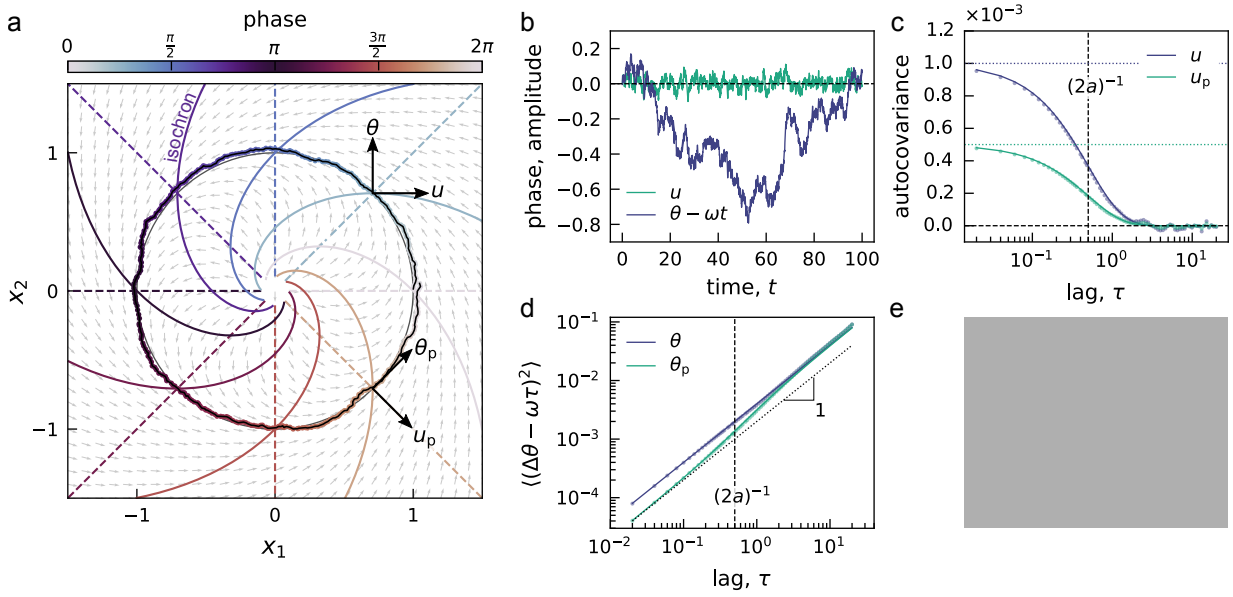


FIG. 2. **Phase–amplitude decompositions of noisy oscillations.** The Stuart–Landau oscillator provides a minimal model illustrating the distinction between asymptotic and projection phase coordinates in the presence of stochastic fluctuations. Simulations were performed with parameters $a = b = 1$, state-space diffusivity $D_x = 0.001$, and sampling interval $\Delta t = 0.02$. (a) State-space representation of a noisy trajectory (black) near the deterministic limit cycle (gray circle). Solid and dashed colored curves denote contours of the asymptotic phase θ (isochrons) and projection phase θ_p, respectively. Arrows indicate the associated phase and amplitude directions. The deterministic vector field is shown in gray. (b) Amplitude u and phase θ associated with the trajectory in (a). The relative phase $\theta - \omega t$ performs a random walk, while the amplitude exhibits bounded fluctuations about zero. (c) Amplitude autocovariances for the asymptotic and projection decompositions. Both decay exponentially with relaxation time $\tau_r = (2a)^{-1}$. (d) Drift-subtracted phase mean-squared displacement, $\langle(\Delta\theta(\tau) - \omega\tau)^2\rangle$, for the asymptotic phase θ and projection phase θ_p. At short times, the projection phase diffuses with slope $2D_x\tau$; at long times, both phases converge to the asymptotic diffusivity $D_\theta = (1 + b^2/a^2)D_x$. The crossover occurs on the relaxation timescale τ_r. (e) Inferred phase diffusivity versus added noise variance. Treating unresolved transverse fluctuations as effective observation noise enables recovery of the asymptotic diffusivity predicted by stochastic phase reduction.

cycle.

While the amplitude coordinates differ only in their parameterization of transverse fluctuations, the corresponding phase coordinates exhibit markedly different stochastic dynamics. The asymptotic phase θ is constructed to eliminate the effects of transverse fluctuations, so that the mean-squared displacement of the relative phase $\theta - \omega t$ grows linearly with time on all timescales (Fig. 2d). In contrast, the projection phase θ_p remains coupled to amplitude fluctuations: transverse displacements modify the instantaneous phase velocity and thereby contribute to phase dispersion until the amplitude relaxes back to the limit cycle. As a result, the relative projection phase $\theta_\mathrm{p} - \omega t$ exhibits a crossover from a short-time diffusivity governed by direct phase fluctuations to the larger asymptotic diffusivity D_θ, with a characteristic crossover time set by the relaxation time τ_r. The enhanced diffusivity at long times reflects both direct phase fluctuations and the cumulative effects of transient amplitude fluctuations.

The distinction between short- and long-time phase dynamics has important implications for inference. Although the latent phase inferred from observations should be interpreted as a projection phase, the phase diffusivity inferred from sufficiently long trajectories can nevertheless be identified with the asymptotic diffusivity predicted by phase reduction theory. This correspondence also highlights a limitation of the reduced phase model, which assumes diffusive phase dynamics on all timescales. As a result, observations sampled on timescales shorter than τ_r contain dynamical information that is not captured by the reduced phase model. Under these conditions, strategies are needed to prevent short-time fluctuations from biasing the inference of the long-time diffusivity.

B. Added observation noise as a regularization strategy

Added observation noise provides a simple mechanism for reducing the influence of short-time fluctuations that are not captured by the reduced phase model. While the reduced phase model provides a compact low-dimensional description of the dynamics, it neglects amplitude fluctuations entirely and is therefore misspecified on timescales shorter than the transverse relaxation time τ_r. The observed signal contains fluctuations arising from both phase and amplitude dynamics, whereas the reduced model represents only phase dynamics. Consequently, if the observation noise is too small, fast amplitude fluctuations are incorrectly interpreted as phase fluctuations, leading to biased estimates of the phase diffusivity. By augmenting the observations with additional noise, a larger fraction of these discrepancies is absorbed into the observation model rather

than the latent phase dynamics, thereby reducing sensitivity to short-time fluctuations.

We implement this regularization by augmenting the observations with additional noise of variance R_{add} and performing inference on the resulting noise-augmented signal. The effective observation noise R_{eff} inferred by the reduced model then reflects the combined effects of measurement noise, added noise, and unresolved amplitude fluctuations. Rather than attributing every deviation from the deterministic waveform to phase dynamics, the inference procedure absorbs a portion of this variability into the observation model, thereby reducing the influence of short-time fluctuations on the inferred phase dynamics. Figure 2e illustrates this idea for the Stuart–Landau oscillator, where the inferred phase diffusivity converges toward the asymptotic value predicted by phase reduction theory as the added noise is increased.

In practice, we determine R_{add} using posterior predictive checks (PPCs) based on the long-time phase dynamics. Specifically, we compare the phase MSD inferred from the smoothing distribution with the posterior predictive distribution of the same statistic generated from trajectories simulated using the inferred phase model. The added noise should be chosen as small as possible while preserving consistency between the inferred phase dynamics and the reduced phase model at long times. Values of R_{add} that are too small lead to biased diffusivity estimates and systematic discrepancies between the inferred and predicted phase dynamics. In contrast, excessively large values unnecessarily discard information about the phase trajectory and increase uncertainty in the inferred quantities. Additional discussion on the choice of R_{add} and its relationship to amplitude fluctuations and transverse relaxation dynamics is provided in the Supplementary Material.

IV. REAL-DATA INFERENCE AND MODEL CRITICISM

To illustrate the inference framework on experimental data, we analyze recordings of the extracellular local field potential (LFP) from the CA1 region of the rat hippocampus obtained during open-field foraging.[16,17] The LFP reflects the aggregate electrical activity of local neuronal populations and exhibits a nearly periodic oscillation at approximately 8 Hz, known as the theta rhythm. The instantaneous phase of this rhythm serves as a temporal reference for neuronal activity and is used to study how the hippocampus encodes spatial information and memory. Consequently, accurate estimation of the stochastic phase dynamics is an important problem in neuroscience.[6,8] Figure 3a shows a segment of a single recording channel sampled at 1250 Hz, corresponding to a minimum time interval of $\Delta t_{\mathrm{min}} = 0.8$ ms. Using this data, we infer a reduced stochastic phase

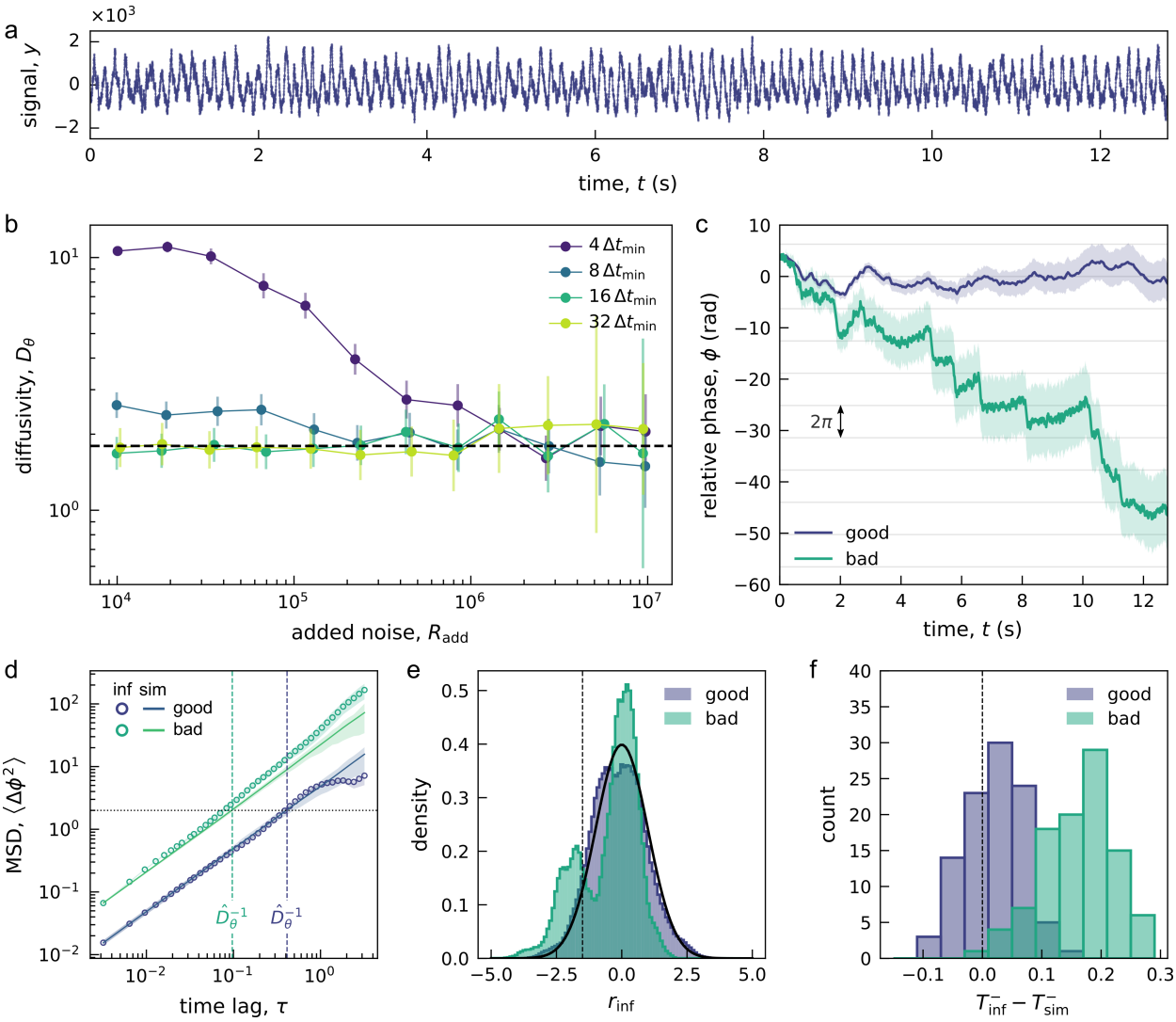


FIG. 3. **Real-data inference and model criticism.** (a) Segment of a hippocampal theta oscillation recorded at 1250 Hz.[16,17] (b) Inferred phase diffusivity D_θ as a function of the added observation noise variance R_{add} for different sampling intervals Δt. (c) Posterior samples of the relative phase, $\phi = \theta - \omega t$, inferred under "good" ($R_{\text{add}} = 10^6$) and "bad" conditions ($R_{\text{add}} = 10^4$) for $\Delta t = 4\Delta t_{\text{min}}$. Solid curves denote the posterior mean, while the shaded regions indicate the interquartile range of the posterior distribution $p(\theta \mid y)$. (d) Drift-subtracted mean-squared displacement of the inferred phase (circles) and simulated phase (solid lines). Vertical dashed lines indicate the characteristic lag $\tau = \hat{D}_\theta^{-1}$ used for the posterior predictive check. (e) Distribution of the scaled phase displacement, $r = \Delta\phi / \sqrt{2 D_\theta \tau}$, evaluated at $\tau = \hat{D}_\theta^{-1}$. The solid curve shows the standard normal distribution, while the dashed line denotes the threshold $z = -1.5$ used to define the lower-tail statistic T^-. (f) Posterior predictive distributions of the discrepancy statistic $T^-_{\text{inf}} - T^-_{\text{sim}}$.

model and apply posterior predictive checks (PPCs) to determine if and when the inferred phase dynamics are consistent with the reduced model.

The inference procedure has two user-specified analysis parameters: the sampling interval Δt and the variance R_{add} of the added observation noise. If the reduced model provides a self-consistent description of the data, the inferred phase dynamics at long times should be insensitive to these choices. Conversely, systematic dependence on either Δt or R_{add} suggests that the assumptions underlying the reduced model are not satisfied. We therefore repeat the inference after thinning the data to larger sampling intervals and augmenting the observations with increasing levels of added noise (Fig. 3b). For sufficiently large sampling intervals ($\Delta t \geq 16\Delta t_{\mathrm{min}}$), the inferred diffusivity is nearly independent of both Δt and R_{add}, clustering about a common value of $D_\theta = XXX$ rad^2/s. In contrast, analyses performed at smaller sampling intervals (e.g., $\Delta t = 4\Delta t_{\mathrm{min}}$) exhibit substantially larger diffusivities for small levels of added noise. As the added noise variance increases, these estimates gradually approach the same consensus value obtained from the more coarsely sampled data. Although this convergence is encouraging, experimental data provide no ground-truth phase trajectory against which the inferred diffusivity can be compared. How then can we determine whether a particular choice of Δt and R_{add} enables a self-consistent description of the data?

To address this question, we seek an internal statistical test capable of determining whether a particular inference is self-consistent with the reduced model. As representative examples, we compare two analyses performed at the same sampling interval, $\Delta t = 4\Delta t_{\mathrm{min}}$, but with different levels of added observation noise: one expected to provide a self-consistent description of the data ($R_{\mathrm{add}} = 10^6$, labeled "good") and another suspected of violating the assumptions of the reduced model ($R_{\mathrm{add}} = 10^4$, labeled "bad"). Figure 3c compares phase trajectories sampled from the posterior distribution $p(\theta \mid y)$ for these two conditions. Qualitatively, the good inference exhibits the stochastic phase wandering expected for diffusive dynamics, whereas the bad inference contains repeated downward jumps of approximately 2π, corresponding to abrupt phase slips. Because the reduced model attributes stochastic phase evolution to diffusion, these slips artificially inflate the inferred diffusivity. These visual observations of phase slips as signatures of model misspecification motivate the development of posterior predictive checks (PPCs) that quantify departures from the reduced phase model.

The PPC asks whether the fitted phase model can reproduce the inferred phase dynamics. To answer this question, we compare two populations of phase trajectories: inferred phase trajectories

sampled from the posterior distribution $p(\theta_{1:N} \mid y_{1:N})$ and simulated phase trajectories sampled from the posterior predictive distribution $p(\tilde{\theta}_{1:N} \mid y_{1:N})$. In practice, we sample model parameters $\{\lambda_s\}$ from the normal approximation to the posterior distribution $p(\lambda \mid y_{1:N})$ for $s = 1, \ldots, S$. For each parameter sample λ_s, we generate paired trajectories for the inferred phase from $p(\theta_{1:N} \mid y_{1:N}, \lambda_s)$ and the simulated phase from $p(\tilde{\theta}_{1:N} \mid \lambda_s)$. As an initial comparison, Figure 3d shows the mean-squared displacement (MSD) of the inferred and simulated phase trajectories. At short lag times, the two agree closely for both the good and bad conditions because the inference procedure is designed to reproduce the observed short-time dynamics. Differences emerge only at longer lag times, where repeated phase slips accumulate into measurable deviations from diffusive behavior. We therefore focus on lag times of order $\tau \sim 1/\hat{D}_\theta$, where the cumulative effects of these rare events are most pronounced while remaining well sampled by the finite data record.

For an ideal drift-diffusion process, the scaled phase displacement

$$r = \frac{\Delta\theta(\tau) - \omega_s \tau}{\sqrt{2D_{\theta,s}\tau}} \tag{10}$$

should follow a standard normal distribution. Figure 3e compares this distribution for the inferred and simulated trajectories for $\tau = 1/\hat{D}_\theta$. The good inference closely follows the expected Gaussian behavior, whereas the bad inference exhibits an excess of large negative displacements arising from the phase-slip events. To quantify these deviations, we define the lower-tail statistic $T^- = \Pr(r < z)$, where $z = -1.5$ is chosen to emphasize rare negative excursions while maintaining adequate sampling statistics. Figure 3f shows the distribution of the difference $T^-_{\text{inf}} - T^-_{\text{sim}}$ over posterior parameter samples. The good inference exhibits no statistically significant discrepancy between inferred and simulated trajectories and therefore passes the posterior predictive check. In contrast, the bad inference consistently produces larger negative tails than predicted by the inferred stochastic phase model, yielding a small posterior predictive p-value and indicating that the reduced phase model is not self-consistent under these conditions.

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These results demonstrate that posterior predictive checks provide a practical criterion for validating reduced stochastic phase models in experimental data. Rather than comparing the inferred phase with an unknown ground truth, the method asks whether the inferred phase dynamics can be reproduced by the inferred model itself. For hippocampal theta oscillations, analyses performed at sufficiently coarse sampling intervals—or equivalently with sufficient effective observation noise—produce statistically self-consistent phase dynamics and converge to a common

estimate of the long-time phase diffusivity. Analyses performed at the native sampling rate with insufficient observation noise instead exhibit systematic phase-slip events that cannot be reproduced by the inferred model and are therefore rejected.

V. COMPARISON TO OTHER METHODS

Compare to other methods.

- FFT analysis
- Hilbert transformation
- State-space EKF
  - specified number of frequencies
  - infer amplitude and phase of each sub-signal
  - amplitudes and phases diffuse?

VI. DISCUSSION

- Asymptotic phase reduction
  - Define "small noise" to neglect higher order contributions
  - Assumes phase and amplitude evolve independently
- Projection phase reduction
  - Define "small noise" to neglect higher order contributions
  - Assumes phase and amplitude evolve independently
    - Failure: phase evolution depends on amplitude (and vice versa)
- Our generative model
  - Assumes phase can be unwrapped?
  - Assumes that phase diffuses with constant diffusivity:

        * Time correlations can break this (even at small noise) - needs multiple relaxation times
        * Phase-dependent diffusivity (but over many cycles only average matters) - needs multiple periods
    - Assumes constant deterministic drift:
        * Failure: amplitude-dependent drift

- Our linearized model

This should synthesize:

- relation to stochastic phase reduction,
- latent-variable inference,
- experimental applicability,
- limitations,
- future extensions.

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DATA AVAILABILITY STATEMENT

AIP Publishing believes that all datasets underlying the conclusions of the paper should be available to readers. Authors are encouraged to deposit their datasets in publicly available repositories or present them in the main manuscript. All research articles must include a data availability statement stating where the data can be found. In this section, authors should add the respective statement from the chart below based on the availability of data in their paper.

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